YEAR 9 MATHS · CHAPTER 7 · MIND MAP

# Geometry



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## 1 Angles and triangles

Triangle angles add to 180°. Angles on a straight line add to 180°, around a point 360°.

Two angles are 50° and 60°: the third is **70°**

Exterior angle = sum of the two opposite interior angles

An isosceles triangle has two equal sides and two equal base angles.

## 2 Parallel lines

Corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

If one angle is 65°, its co-interior angle is **115°**

**Watch out:** these rules only work when the lines are parallel.

## 3 Quadrilaterals and polygons

Interior angle sum of an $n$-sided polygon: $(n-2)\times 180^\circ$.

Quadrilateral: 360°

Hexagon: $4\times 180^\circ = 720^\circ$; each angle of a regular hexagon is 120°

## 4 Congruent triangles

Congruent means exactly the same shape and size (≅). Tests: SSS, SAS, AAS, RHS.

SSS: all three sides equal

SAS: two sides and the angle between them

RHS: right angle, hypotenuse and one side

**Watch out:** AAA is not a congruence test: same angles can be different sizes.

## 5 Using congruence in proof

List the equal parts with a reason for each, name the test, then conclude.

ABCD is a parallelogram. Prove △ABC ≅ △CDA.

AB = CD and BC = DA (opposite sides of a parallelogram)

AC is common, so △ABC ≅ △CDA (SSS)

## 6 Enlargement and similar figures

Scale factor $= \dfrac{\text{image length}}{\text{original length}}$. Angles stay the same.

A 4 cm side enlarged by a scale factor of 2.5 becomes **10 cm**

Similar figures: same shape, sides in the same ratio.

## 7 Similar triangles

Tests: AAA, SSS (ratio), SAS (ratio), RHS (ratio). Match the sides in order.

Matching sides 4 and 6: scale factor $\frac{6}{4} = 1.5$

Side 5 matches $x$: $x = 5\times 1.5 = 7.5$

## 8 Proving and applying similar triangles

Show the triangles are similar, then use the scale factor.

A 2 m pole casts a 3 m shadow. A tree's shadow is 12 m at the same time.

Height $= 2\times\frac{12}{3} = 8$ m

**Watch out:** write the vertices in matching order: △ABC ||| △DEF.